Boshi Wang

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Education

ShanghaiTech University Shanghai, China
Major in Computer Science Sep. 2016 - Present

- GPA: 3.76/4.0
- Awards: Dean's Scholarship (2016), Excellence Scholarship (2016-2017), Dean's Scholarship (2017-2018)
- Selected Coursework:

Applied Algebraic Geometry	(undergraduate)
Convex Optimization	(graduate)
Matrix Analysis	(graduate)
Introduction to Algorithmic Game Theory	(graduate)
Subspace Learning (Fall 2019)	(graduate)

Publication

* denotes equal contribution

Published

- Z. Tu, **B. Wang**, T. Duan, and F. Gao. Pattern-learning-based Noise Elimination Algorithm in Photoacoustic Sensing and Imaging. In *40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC'18)*, Honolulu, HI, USA, on July 17-21, 2018.

Under Review

- Y. Guo*, **B. Wang***, D. Hughes, K. Sycara. Designing Context-Sensitive Norm Inverse Reinforcement Learning Framework for Norm-Compliant Autonomous Agents. *Submitted to ICRA 2020.*

In Progress

- **B. Wang**, M. Tsakiris, Homomorphic Sensing with a Union of Subspaces as Input Space. Link: blablabla

Research Experience

Research Interest: **Robotics, Machine Learning, Artificial Intelligence, Subspace Learning**

Advanced Agent-Robotics Technology Lab, Carnegie Mellon University Pittsburgh, USA
Undergraduate Research Intern, supervised by Prof. *Katia Sycara* (participant of Robotics Institute Summer Scholars (RISS) program) Jun 2019 - Sep 2019

- Developed Context-Sensitive Norm Inverse Reinforcement Learning (CNIRL), a new IRL framework which can enable an agent to learn to not only perform domain tasks but also act

according to various social norms and legal/moral rules from demonstration trajectories under large number of environmental contexts.

Manolis Research Group, ShanghaiTech University Shanghai, China
Undergraduate Research Assistant, supervised by Prof. *Manolis C. Tsakiris* Feb 2019 - Present

- Successfully extended all the existing theorems in Homomorphic Sensing (a new research topic proposed by Prof. Tsakiris in an ICML'19 paper) into the case where the input space is a union of finitely many linear subspaces instead of a single linear subspace.
- Currently conducting theoretical analysis on solving linear regression with paritially permuted data with Dual Principle Component Pursuit (DPCP)/Sparse Subspace Clustering (SSC), and also on further extending Homomorphic Sensing into the case where the input space is an arbitrary algebraic variety defined by homogeneous polynomials.

Hybrid Imaging System Laboratory, ShanghaiTech University Shanghai, China
Undergraduate Research Assistant, supervised by Prof. Fei Gao Jan 2018 - Nov 2018

- Developed an algorithm that can learn from historical data about the pattern of a periodic noise signal, then efficiently detect and eliminate the noise in newly-collected photoacoustic signals thus achieve image quality enhancement. (Paper accepted by *EMBC'18*)
- Conducted research on using denoising autoencoders (DAE) in photoacoustic signal noise reduction, and found an interpretation of the encoder-decoder mechanism which has a very profound meaning in signal-processing context and is related to the Spectral Bias of Neural Networks. The denoising behavior largely outperforms existing methods under low signal-to-noise ratio.

Mobile Autonomous Robotic Systems Lab, ShanghaiTech University Shanghai, China
Undergraduate Research Assistant, supervised by Prof. *Sören Schwertfeger* Jun 2017 - Dec 2017

- Successfully developed and implemented an algorithm which can enable a robot to localize itself based completely on visual information of Aruco markers in surrounding environment collected by a camera on the robot. The work contributes to a bigger project in the lab about Simultaneous Localization and Mapping.

Teaching Experience

SI140: Probability and Mathematical Statistics, ShanghaiTech University Shanghai, China
Teaching Assistant

Internship

Chinese Software Developer Network Corporation (CSDN) Beijing, China
Intern of System Developer Jun 2018 - Aug 2018

- Optimized the keyword matching and filtering algorithm of the blog recommendation system.
- Developed an efficient method for clustering search queries of blogs and implemented efficient parallel computing using Spark.

Other Selected Projects

Deep Q-learning based AI agent for Beating the Landlord
Course project of *CS181: Artificial Intelligence* in ShanghaiTech University, Fall 2018

- Developed a Deep Q-learning framework which learns to play a popular 3-player card game in China called "Beating the Landlord" through three agents' self-play.
- Although the trained agent still can't beat human players, it largely outperforms greedy agents which are used in most online game platforms.

Miscellaneous

Programming Skills: Python, C/C++, Matlab, Rust, MIPS assembly

Tools: CVX, Tensorflow, Pytorch, Robot Operating System (ROS)

Sports: Billiards, Badminton, Ping-Pong, Soccer

Other Interests: Guitar, Dota2, a big fan of Japanese animation *Ghost in the Shell*.